

Yeaseen Arafat

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RESEARCH INTERESTS

Broadly interested in Systems and Software Security, focusing on automated software testing, program analysis, compiler design, and generative AI for fuzzing, program generation, and bug triage.

EDUCATION

🎓 The University of Utah

Ph.D. in Computer Science

M.S. in Computer Science

Aug 2023 – Present

Aug 2023 – Dec 2025 (Expected)

🎓 Bangladesh University of Engineering and Technology (BUET)

B.Sc. Engg. in Computer Science and Engineering

Feb 2015 – April 2019

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

May 2024 – Present

FuTURES³ Lab, The University of Utah.

Utah, USA

- Built a GenAI-assisted program generator for multi-language corpora, producing diverse test cases used to fuzz code translators across C→Rust/Go/Zig, JS6+→JS5, and Go→Haxe pairs.
- Designed and implemented **TeTRIS**, a general-purpose mutational fuzzing framework that uncovers semantic translation bugs in source-to-source compilers (e.g., C2Rust, CxGo, Go2Hx).
- Developed a cross-language parser that normalizes C/Go/Haxe into a unified Abstract Syntax Tree with full scope & type resolution, enabling semantics-preserving program mutations.

Lecturer

June 2019 – July 2023(On Leave)

Department of Computer Science and Engineering, Southeast University

Dhaka, Bangladesh

- Taught core undergraduate CS courses and supervised thesis/final year projects in machine learning and data analysis, while contributing to curriculum development.

Undergraduate Student Researcher

Apr 2018 – Apr 2019

Bangladesh University of Engineering and Technology

- Designed an ML-based intrusion detection system in the MAC layer of Network Simulator-2, using statistical features (e.g., packet ratios, inter-arrival times) to differentiate malicious and benign nodes.

OPEN SOURCE CONTRIBUTIONS

- **Bug Discovery:** Discovered and reported **65+** developer-confirmed previously-unknown bugs across projects such as `c2rust`, `Zig translate-c`, `cxgo`, `Go compiler`, and `go2hx`. [FuTURES³ Lab's Bug Tracker]
- **Upstream Contributions:** Contributed patches through merged PRs to major open-source projects, including `llvm-project`, `c2rust`, `Polyglot` and `cxgo`. [details]
- **Responsible Disclosure:** Reported a critical key exposure vulnerability in a mobile banking app. [details]

PUBLICATIONS

Conference:

- **Yeaseen Arafat**, Stefan Nagy, “Testing Transpilers via Fuzzing”, *under review*.
- **Yeaseen Arafat**, Stefan Nagy, “TeTRIS: General-purpose Fuzzing for Translation Bugs in Source-to-Source Code Transpilers”, Annual Computer Security Applications Conference (ACSAC 2025), Hawaii, USA. [pdf] [code]
- **Yeaseen Arafat**, Kazi Samin Yeaser, Ashikur Rahman, Arnab Dasgupta, “A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks”, International Conference on Networking, Systems and Security (NSysS 2020), December 22-24, 2020, Dhaka, Bangladesh. [doi] [code]

TALKS AND PRESENTATIONS

- **Poster: Yeaseen Arafat**, Stefan Nagy, “TeTRIS: Fuzzing for Code Translation Bugs in Source-to-Source Code Transpilers”, Kahlert School of Computing Graduate Visit Day, February 21–22, 2025, University of Utah. [[poster](#)]
- **Paper: Yeaseen Arafat**, Kazi Samin Yeaser, Ashikur Rahman, Arnab Dasgupta, “A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks”, International Conference on Networking, Systems and Security (7th NSysS 2020). [[talk](#)] [[slide](#)]

TECHNICAL PROJECTS

CyberShield-Spectrum | *Virtualized Security Lab*

 [GitHub](#)

- Built and tested a virtual enterprise network with VPN, firewall, IDS, and jump server configurations.
- Performed penetration testing and threat hunting using Snort, Volatility3, and FTK Imager.

KerFuzz: Linux Kernel Fuzzer | *C, Bash, GDB, Syzkaller*

 [GitHub](#)

- Designed a syscall-level kernel fuzzing workflow using Syzkaller to discover bugs in the Linux kernel.
- Instrumented and analyzed faulting executions using GDB and custom Bash scripts for postmortem triage.

EduCComp | *Flex, Bison, YACC*

 [GitHub](#)

- Built a compiler for a C-subset, including Symbol Table, Lexical, Syntax, and Semantic Analysis.
- Implemented an Intermediate Code Generator, translating source code into x86 assembly.

Machine Learning projects | *Python*

 [GitHub](#)

- Implemented ML algorithms from scratch, including AdaBoost with Decision Trees, Perceptron, and ANN.
- Built recommender (Matrix Factorization), unsupervised models (PCA, GMM), and Channel Equalizer.

IInstagram | *MERN Stack*

 [GitHub](#)

- Built an Instagram-like web app with backend APIs (Node.js/Express), JWT auth, and image uploads (Cloudinary).
- Implemented password recovery with Twilio email integration, along with React Hooks, Router, and Context API for state management and group chat using Socket.IO.

AWARDS & SCHOLARSHIP

- **Graduate Research Fellowship**, University of Utah (Aug–Dec 2023)
- **University Merit Scholarship**, BUET (2015–2019)
- **Junior Government Scholarship**, Chittagong Education Board, Bangladesh (2009)

OUTREACH ACTIVITIES

- **Artifact Evaluation Committee Member:** [ACM CCS 2024](#), ACSAC 2025
- **Graduate Teaching Assistant:** CSE-4480 (Computer Networks), University of Utah (Spring 2024). Supported 170+ students via office hours, grading, and final evaluations.
- **Conference Attendee:** KubeCon+CloudNativeCon North America 2024 — Highlights from KubeCon 2024
- **Official Collaborator:** Invited by project maintainers of Go2Hx for ongoing contributions to bug discovery.
- **Organizing Committee Member:** ACM ICPC ASIA Regional, Dhaka Site 2019
- **Member:** ACM; Cybersecurity Club, University of Utah [[utahsec](#)]

TECHNICAL SKILLS

Programming & Systems: C/C++, Go, Rust, Python, Java, Assembly (x86, MIPS), Wasm, Bash

Security & Fuzzing: AFL++, Burp Suite, Wireshark, sqlmap, Snort, Ghidra, Pwndbg

Compilers & Analysis: LLVM, Flex/Bison/YACC, Nachos, Make, Static Analysis

Machine Learning: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy

Web & DevOps: React, Node.js, Express, Docker, GitHub Actions, GraphQL, REST API

REFERENCES

Dr. Stefan Nagy

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Kahlert School of Computing
The University of Utah
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Dr. A.K.M. Ashikur Rahman

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